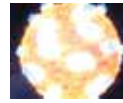




Water Vapor found on Ganymede

Scientists have found evidence of water vapor on Jupiter's moon - Ganymede's surface. <https://digit.in/sep21-31>



Witnessing a supernova

A billion years ago a star exploded in the constellation of Cancer, and that light has reached us now. <https://digit.in/sep21-32>

Credit: NASA

SEARCHING FOR MARTIANS

If ancient life inhabited Mars billions of years ago, Perseverance is in just the right place to find signs of it

Aditya Madanapalle | aditya@digit.in

In 2018, astrobiologists were in a tizzy because of a paper by planetary scientists that indicated that there might be subsurface lakes deep beneath the Martian south pole. The findings were based on radar probes

conducted by Mars Express, an orbiter by the European Space Agency. This finding was exciting because where there is water, there is life, at least on Earth. While at the surface any water is likely to quickly vaporise in the low density atmosphere, deep within the surface, the conditions could be right for maintaining liquid water. These subsurface lakes could be the habitat for microbes. Now, researchers have reanalysed the data from the radar probes and conducted experiments on cold labs on Earth. Three papers published recently indicate that the detections were caused because of clays and not subsurface lakes. The supposed lakes were too close to the surface and too cold for the water to be liquid, even after

factoring the presence of perchlorates or brine, that keeps water in the liquid state at colder temperatures. The analysis of over 15 years of radar data turned up dozens of additional patches similar to the one found in 2018, but all of these are now believed to be deposits of clay. One of the three papers suggest many potential materials that could have caused the radar probes to flag the presence of water, including minerals with high metal content and frozen saltwater. Another team of researchers investigated a group of clays known as smectites that are common on Mars. These smectites look like ordinary rock, but were formed by liquid water billions of years ago on the planet. The team cooled down smectites on Earth in a cold